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Large Language Model Guided Progressive Feature Alignment for Multimodal UAV Object Detection

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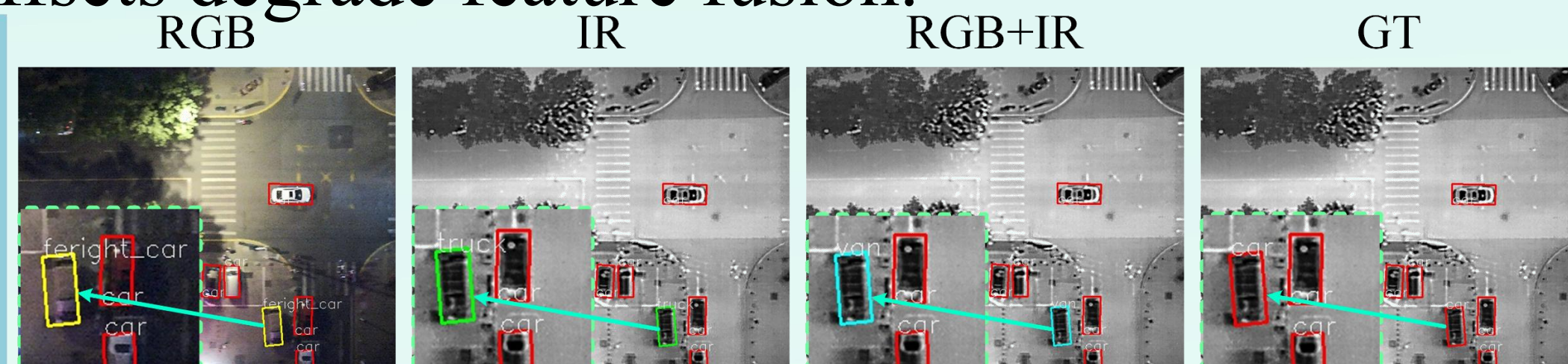
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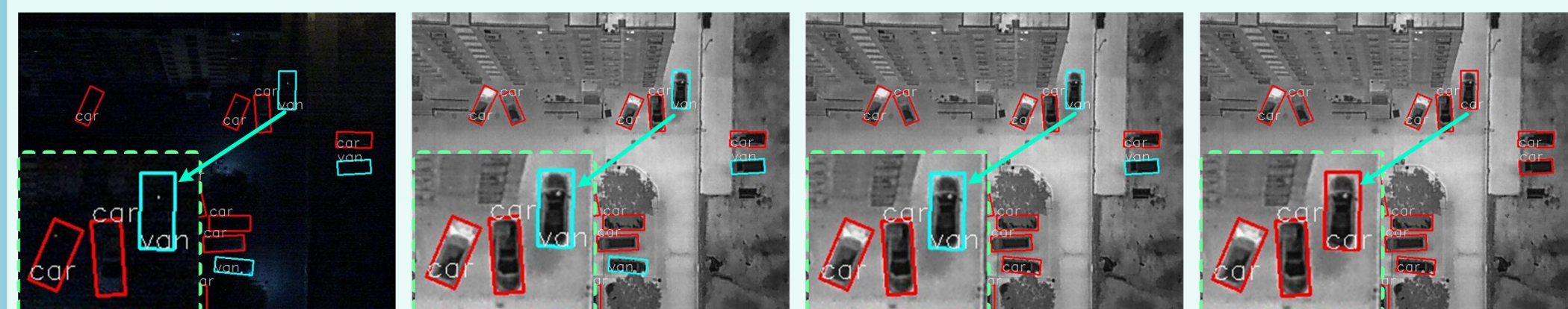
Introduction

Challenges:

- (1) Cross-modal semantic differences cause category confusion. (2) RGB-IR spatial position offsets degrade feature fusion.

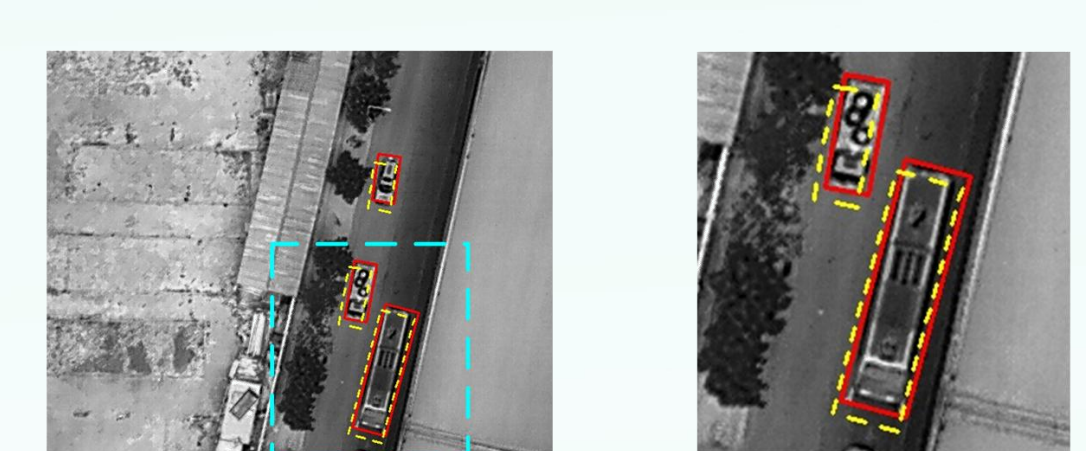
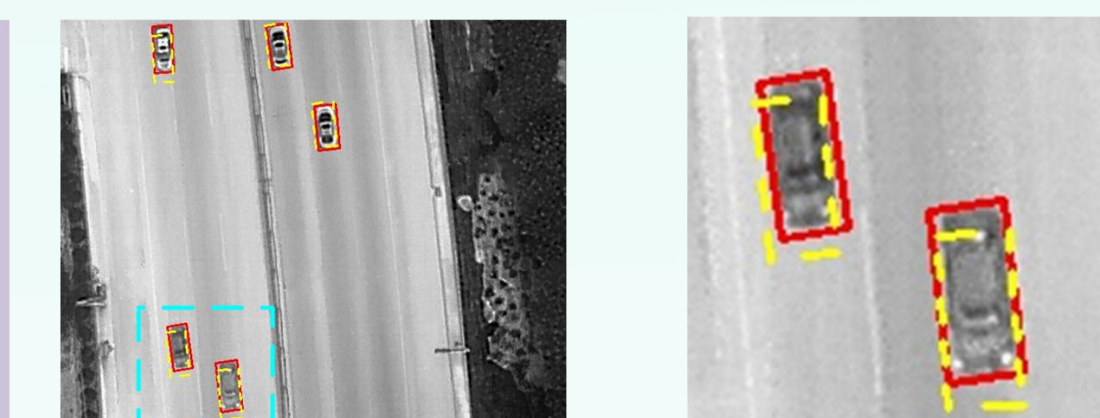


(a) Semantic ambiguity in fusion caused by cross-modal semantic differences.



(b) Semantic differences between visual representations and category labels.

Car Van Truck Freight car Bus

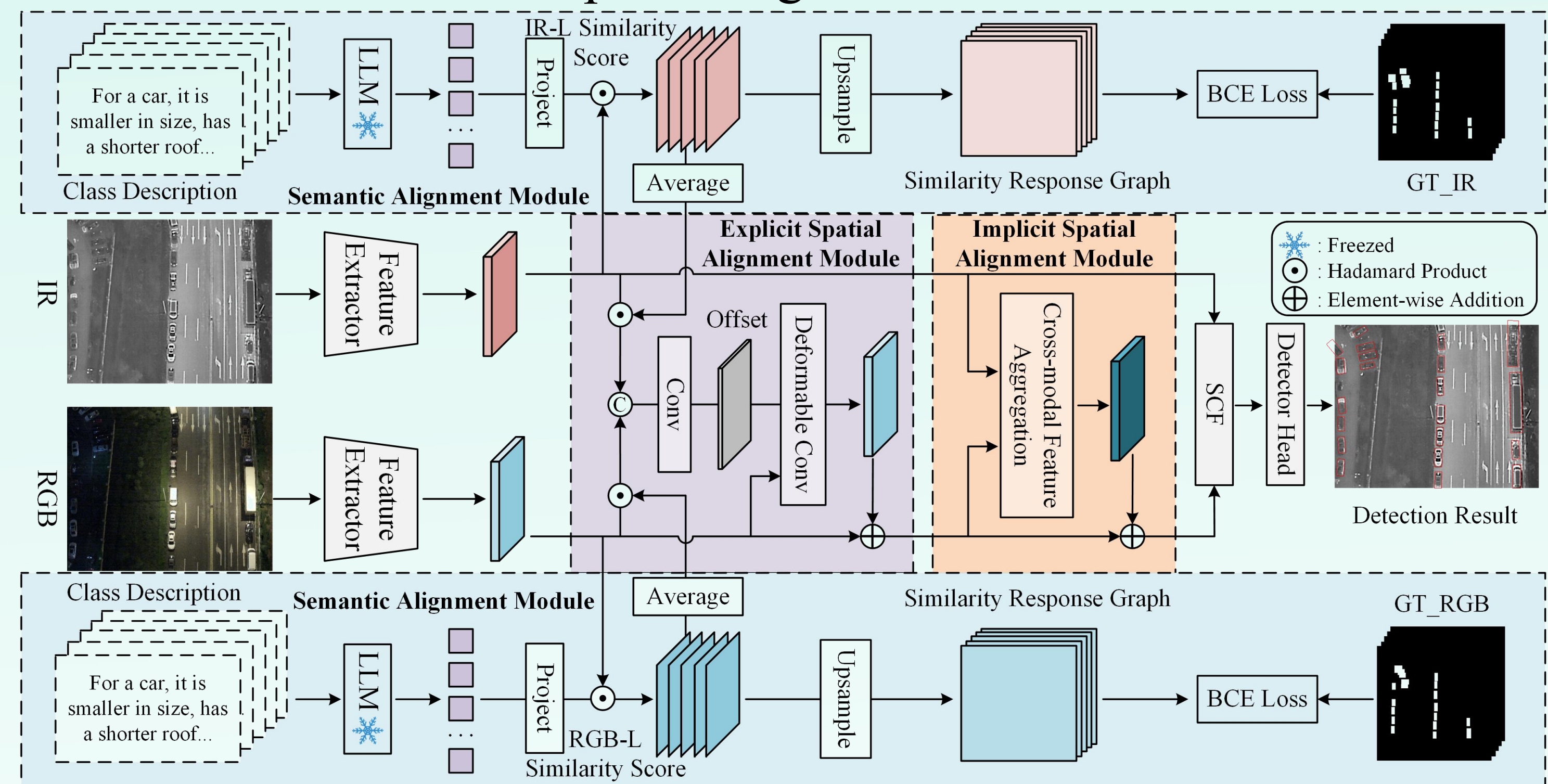


(c) Spatial positional differences between different modalities.

True value of IR True value of RGB

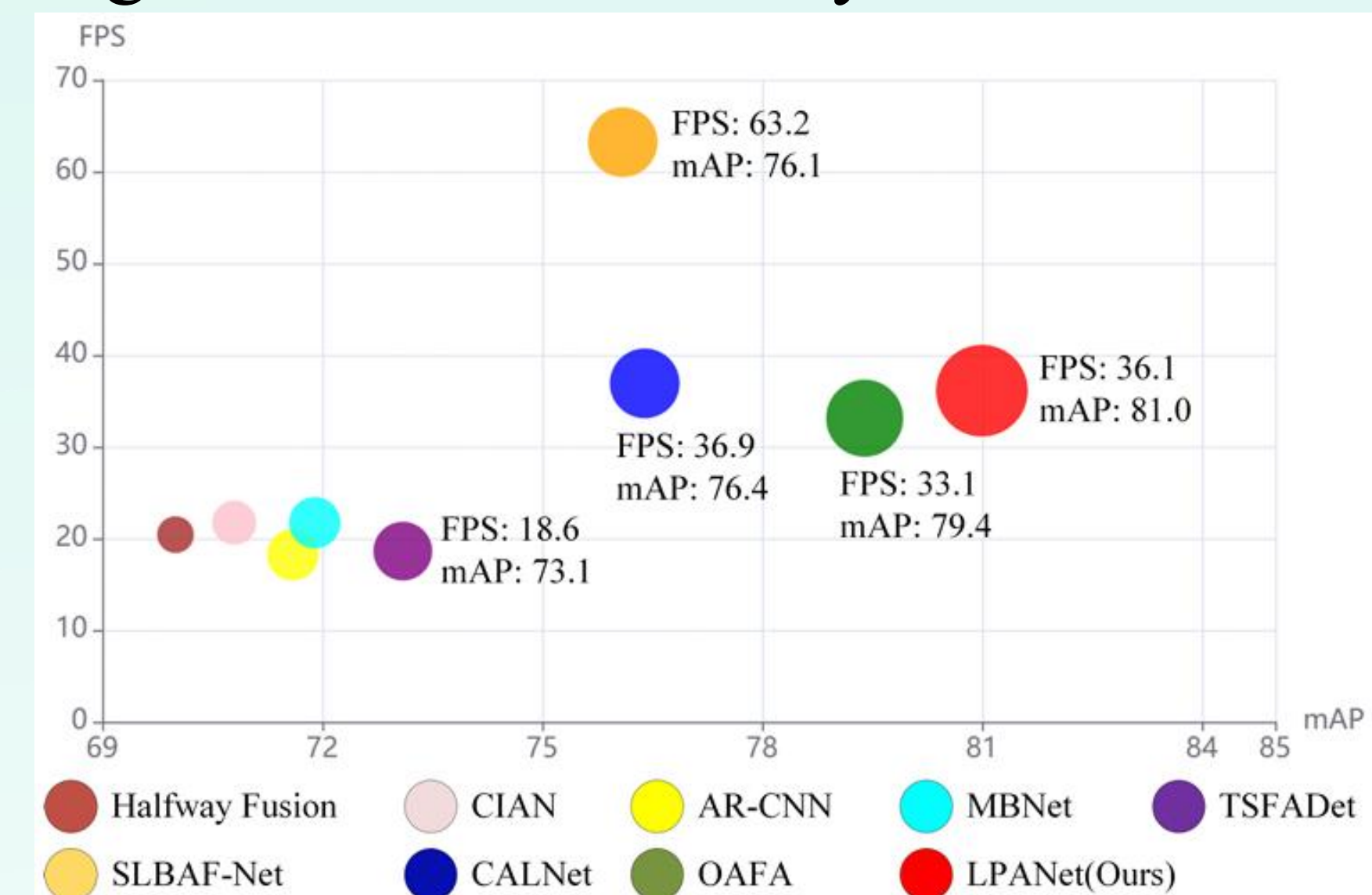
Method

We propose a LLM guided Progressive Feature Alignment Network to achieve semantic and spatial alignment between modalities.



Analysis

Significant gains over baseline with high inference efficiency maintained.



Method	Car	Truck	F.car	Bus	Van	mAP _{0.5}	FLOPs
Baseline	90.3	73.7	68.7	89.7	59.7	76.4	171.53G
+SAM	90.2	75.9	72.0	89.8	61.3	77.8	171.58G
+ISM	90.4	78.0	74.6	90.0	63.2	79.3	184.10G
+ESM	90.4	80.6	76.3	90.1	67.5	81.0	184.63G

Result

Detectors	DroneVehicle Val Set						DroneVehicle Test Set					
	Car	Truck	Freight car	Bus	Van	mAP _{0.5}	Car	Truck	Freight car	Bus	Van	mAP _{0.5}
UA-CMDet [26]	-	-	-	-	-	-	87.5	60.7	46.8	87.1	38.0	64.0
Halfway Fusion [41]	90.1	62.3	58.5	89.1	49.8	70.0	-	-	-	-	-	-
CIAN [42]	90.1	63.8	60.7	89.1	50.3	70.8	-	-	-	-	-	-
AR-CNN [13]	90.1	64.8	62.1	89.4	51.5	71.6	-	-	-	-	-	-
MBNet [25]	90.1	64.4	62.4	88.8	53.6	71.9	-	-	-	-	-	-
TSFADet [14]	89.9	67.9	63.7	89.8	54.0	73.1	89.2	72.0	54.2	88.1	48.8	70.4
C ² Former [40]	-	-	-	-	-	-	90.2	68.3	64.4	89.8	58.5	74.2
SLBAF-Net [43]	90.2	72.0	68.6	89.9	59.9	76.1	-	-	-	-	-	-
CALNet [9]	90.3	73.7	68.7	89.7	59.7	76.4	90.3	76.2	63.0	89.1	58.5	75.4
OFA [15]	90.3	76.8	73.3	90.3	66.0	79.4	-	-	-	-	-	-
DPDETR [27]	90.3	78.2	75.7	90.1	64.9	79.8	-	-	-	-	-	-
GAGTDet [7]	90.8	69.7	66.3	90.5	55.6	74.6	-	-	-	-	-	-
Ours	90.4	80.6	76.3	90.1	67.5	81.0	90.4	78.0	65.0	89.5	65.4	77.7

